

Time Tracking Application Technical Specification

Version 3.0 - Final Specification

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System Overview

A multi-user time tracking application with Google authentication, role-based access control, hierarchical task selection, invoice generation with PDF export, and comprehensive administrative management capabilities for a law firm's document review operations.

User Roles & Permissions

1. Employee (Reviewer)

- **Access Level:** Most restricted
- **Capabilities:** Time tracking only
- **Authentication:** Google account (company-provided)

2. Administrative Assistant

- **Access Level:** Mid-level
- **Capabilities:** Client and task management
- **Authentication:** Google account

3. Administrator

- **Access Level:** Full system access
- **Capabilities:** All system operations
- **Authentication:** Google account

Employee Interface

Authentication & Landing

- Single Sign-On via Google OAuth
- Upon login, immediately displays time tracking interface
- No navigation menu or access to other sections
- ONLY sees the time tracker - nothing else

Time Tracking Workflow

1. Client Selection

- Dropdown showing only clients assigned to this employee
- Required first step before tracking time

2. Task Type Selection (Hierarchical)

First Dropdown: Main Task Category

- Filtered by employee's review level (1, 2, or 3)
- Level 1 employees see only Level 1 main tasks
- Level 2 employees see only Level 2 main tasks
- Level 3 employees see only Level 3 main tasks

Second Dropdown: Sub-Task (Conditional)

- Only appears if the selected main task has sub-items
- Dynamically populated based on main task selection
- Shows all sub-tasks under the selected main task
- If main task has no sub-items, this dropdown is hidden/disabled

3. Manual Description Field

- Text area/input field for additional notes
- Allows employee to describe work that doesn't fit dropdown options
- Optional but available for every time entry

4. Timer Controls

- **Start:** Begin tracking time on selected task
- **Pause:** Temporarily stop timer (for breaks, bathroom, etc.)
- **Stop:** Completely end current task

- Can resume after pause
- Can start new task after stop

Administrative Assistant Interface

Navigation

- Has access to administrative menu/sidebar
- Can navigate between different admin sections

Client Management

- Add new clients to the system
- Edit existing client information
- Activate/deactivate clients

Task Management

- Add new main task categories
- Add sub-tasks under main categories
- Edit existing tasks and sub-tasks
- Assign review levels to tasks
- Enable/disable tasks

Administrator Interface

■ CRITICAL: Admin Navigation & Testing

Admin Menu Bar / Sidebar (NOT visible to employees):

The administrator has a persistent navigation menu with the following sections:

- **Dashboard** - Overview, statistics, recent activity
- ■ **View as Employee** - Switch to employee interface for testing (can switch back)
- **Users & Permissions** - Manage employees, roles, and permissions
- **Clients** - Client management and assignments
- **Tasks** - Task hierarchy management (main tasks and sub-tasks)
- **Time Entries** - View and manage all time entries
- **Invoices** - Generate, edit, and export invoices
- **Rates** - Manage hourly rates by review level
- **Company Settings** - Logo, address, invoice templates, etc.

"View as Employee" Mode:

When the administrator clicks "View as Employee", the interface switches to show exactly what a regular employee sees (just the time tracker). However, the admin navigation menu remains accessible (perhaps as a floating button or persistent top bar) so they can click back to "Admin Mode" at any time.

This allows the administrator to:

- Test the employee experience without logging out
- Verify that task filtering works correctly
- Ensure employees can only see their assigned clients
- Switch back to admin functions instantly

1. Users & Permissions Management ■ NEW

User List View:

- Table showing all users with columns:
 - Name
 - Email
 - Role (Employee / Admin Assistant / Administrator)
 - Review Level (for employees: 1, 2, or 3)
 - Status (Active / Inactive)
 - Assigned Clients (count)
 - Actions (Edit / Deactivate / Delete)

Add/Edit User Form:

- Email (Google account)
- Name
- Role dropdown: Employee / Administrative Assistant / Administrator
- Review Level (if Employee): 1, 2, or 3
- Status: Active / Inactive

Bulk Operations:

- Select multiple users and:
 - Assign to client(s)
 - Change review level
 - Activate/deactivate

2. Client Assignment Management

Two-Way Assignment Interface:**Option A: From Client Perspective**

- Select a client
- See list of all employees
- Check boxes next to employees who should have access to this client
- Save assignments

Option B: From Employee Perspective

- Select an employee
- See list of all clients
- Check boxes next to clients this employee should access
- Save assignments

3. Task Hierarchy Management ■ EXPANDED

Task Management Interface:

Administrator needs full CRUD (Create, Read, Update, Delete) capabilities for tasks and sub-tasks.

Main Tasks View:

- Hierarchical tree view or table showing:
 - Task description
 - Review level (1, 2, or 3)
 - Number of sub-tasks

- Status (Active / Inactive)
- Actions (Edit / Add Sub-Task / Delete)

Add/Edit Main Task Form:

- Task description (text field)
- Review level: 1, 2, or 3
- Display order (for sorting)
- Status: Active / Inactive
- Save button

Sub-Tasks Management:

When viewing/editing a main task, show list of sub-tasks with:

- Sub-task description
- Display order
- Status
- Actions (Edit / Delete / Reorder)
- **Add Sub-Task** button within each main task
- Drag-and-drop reordering (optional but recommended)
- Inline editing for quick changes

Bulk Import Feature (Recommended):

- Upload Excel file with task structure
- Preview imported tasks before saving
- Option to replace all tasks or merge with existing
- Export current tasks to Excel for backup/editing

4. Time Entry Review & Management

- View all time entries across all employees
- Filter by: client, employee, date range, task/sub-task, status
- Edit time entries if corrections are needed
- Delete invalid entries
- Export to CSV/Excel

5. Invoice Generation & Editing

Invoice Creation:

1. Select client
2. Select date range
3. System generates editable invoice draft

Invoice Editor - Split View:

Left Panel: Detailed time entries

- Employee name
- Date
- Main task
- Sub-task (if applicable)
- Manual description
- Original hours (read-only)
- Adjusted hours (editable field)
- Hourly rate
- Line total
- Delete button

Right Panel: Live summary (updates in real-time)

- Subtotal by review level
- Total hours
- Total amount
- Adjustment summary (total hours removed/added)

Top Action Bar:

- Save Draft button
- Preview button (shows PDF preview)
- Export to PDF button (primary action)
- Finalize Invoice button (locks invoice)

6. PDF Export & Company Branding

Invoice Header:

- Company logo (upload capability)
- Company name
- Address
- Contact info (phone, email)
- Website, Tax ID (optional)

Invoice Body:

- Invoice number (auto-generated)
- Invoice date
- Client information (Bill To)
- Service date range
- Detailed line items table
- Totals and subtotals

Invoice Footer:

- Payment terms
- Bank details / payment methods
- Thank you message
- Page numbers

7. Rate Management

- Set hourly rates for Level 1, 2, and 3
- Modify rates (with effective date)
- View rate history
- Set default rate for new employees

Testing Strategy & Admin Navigation

Problem:

Employees only see the time tracker with no navigation menu. How does the administrator test the employee experience?

Solution: View Mode Toggle

Implementation:

1. Admin Menu Bar (Always visible to admin):

- Horizontal top bar or left sidebar
- Contains navigation links to all admin sections
- Includes "View as Employee" toggle button

2. When Admin clicks "View as Employee":

- Interface switches to employee view (just time tracker)
- Admin menu remains accessible (e.g., collapsed to icon or small badge)
- Badge/icon says "Exit Employee View" or "Back to Admin"

3. Testing Different Employee Scenarios:

- Admin can temporarily "assume" an employee's identity for testing:
 - Select employee from dropdown in admin menu
 - See exactly what that employee sees (their clients, their tasks)
 - Verify permissions work correctly
- Return to normal admin view anytime

Visual Example:

Normal Admin View:



Employee View Mode (Admin Testing):



■ (what employee sees) ■



Response	Percentage
Doing a good job	55%
Not doing a good job	44%

Regular Employee View (No admin access):

[illegible]

- [Logo] ■ ← No menu, just branding



■ Time Tracker Interface ■

■ (ONLY this - nothing else) ■



Response	Percentage
Doing a good job	55%
Not doing a good job	44%

Data Model Requirements

Users Table

- User ID (Google account ID)
- Email
- Name
- Role (Employee/Admin Assistant/Administrator)
- Review Level (1, 2, 3) - for employees only
- Status (active/inactive)
- Created date
- Last login

Clients Table

- Client ID
- Client name
- Contact information
- Billing address
- Status (active/inactive)
- Created date
- Created by (user ID)

Client Assignments Table

- Assignment ID
- Employee ID
- Client ID
- Assigned date
- Assigned by (admin user ID)

Main Task Categories Table

- Main task ID
- Task description (full text)
- Review level (1, 2, or 3)
- Has sub-tasks (boolean flag)
- Display order
- Status (active/inactive)
- Created date
- Modified date

Sub-Tasks Table

- Sub-task ID
- Main task ID (foreign key)
- Sub-task description
- Display order
- Status (active/inactive)
- Created date
- Modified date

Time Entries Table

- Entry ID
- Employee ID
- Client ID
- Main task ID
- Sub-task ID (nullable)
- Manual description/notes
- Start timestamp
- End timestamp (null if currently running)
- Paused duration (accumulated pause time in seconds)
- Total duration (calculated in seconds)
- Status (in-progress/paused/completed)
- Created date

Hourly Rates Table

- Rate ID
- Review level (1, 2, 3)
- Rate amount (decimal)
- Effective date
- End date (null if current)
- Created by (admin user ID)
- Created date

Invoices Table

- Invoice ID
- Invoice number (formatted string, auto-generated)
- Client ID
- Date range start

- Date range end
- Generated date
- Generated by (admin user ID)
- Status (draft/finalized/sent)
- Total hours
- Original total amount
- Adjusted total amount
- PDF generated (boolean)
- PDF file path (if saved)

Invoice Line Items Table

- Line item ID
- Invoice ID
- Time entry ID (reference)
- Employee name
- Date performed
- Main task name
- Sub-task name (nullable)
- Manual description
- Original hours
- Adjusted hours
- Hourly rate applied
- Review level
- Line total
- Adjustment notes

Company Settings Table

- Setting ID
- Company name
- Company logo (file path or URL)
- Address line 1
- Address line 2
- City
- State/Province
- Postal code
- Country
- Phone number
- Email

- Website
- Tax ID / Business number
- Invoice footer text
- Payment terms
- Payment instructions
- Bank details
- Brand primary color
- Brand secondary color
- Invoice template style
- Last updated
- Updated by (admin user ID)

UI/UX Requirements

Role-Based Interface Design

Employee Interface:

- Minimal, focused design
- Only time tracker visible
- No navigation menu
- Simple top bar with logo and logout

Admin/Admin Assistant Interface:

- Full navigation menu (sidebar or top bar)
- Access to all relevant admin sections
- Dashboard with statistics
- "View as Employee" mode toggle

Navigation & Testing Features

- Admin menu always accessible (even in employee view mode)
- Quick user/role switcher for testing (admin only)
- Breadcrumb navigation in admin sections
- Clear visual distinction between modes

Task Management Interface Requirements

- Tree/hierarchical view of tasks and sub-tasks
- Inline editing for quick updates
- Drag-and-drop reordering
- Bulk operations (activate/deactivate multiple tasks)
- Search/filter functionality
- Import/export Excel capability

Implementation Notes

Permission Checking

Every page/route in the application must check:

- User is authenticated (valid Google account)
- User has the correct role for this section
- User has active status

Example Permission Logic:

Time Tracker page:

- Any authenticated user can access
- Shows only their assigned clients
- Shows only tasks for their review level

Invoice page:

- Only Administrator role
- Redirect employees to time tracker

Task Management page:

- Administrator and Admin Assistant
- Employees get access denied

View Mode Implementation

When admin enters "View as Employee" mode:

- Store admin's actual identity in session
- Set a view_mode flag: 'admin' or 'employee'
- If view_mode='employee', render employee interface
- Always show admin menu badge/button
- On exit, restore normal admin interface

Security Considerations

- Never trust client-side role/permission data
- Always verify permissions server-side
- Log all admin actions (audit trail)
- Prevent employees from accessing admin routes even if they guess the URL
- Session timeout after inactivity

MVP Features (Phase 1)

Essential features to launch:

1. Google authentication with role-based access
2. User management (add users, assign roles and review levels)
3. Client management (add clients)
4. Client assignment (assign clients to employees)
5. Task management interface (add/edit main tasks and sub-tasks)
6. Employee time tracking with hierarchical task selection
7. Manual description field
8. Admin "View as Employee" mode for testing
9. Basic invoice generation with editing
10. PDF export with company header/footer
11. Company settings (logo, address, etc.)
12. Hourly rate management by review level

Future Enhancements (Phase 2+)

1. Email invoices directly from system
2. Multiple invoice templates
3. Reporting and analytics dashboard
4. Export time entries to Excel
5. Email notifications (invoice sent, time entry reminders)
6. Mobile responsive design or mobile app
7. Approval workflows
8. Bulk task import from Excel
9. Task templates
10. Audit log / change history
11. Payment tracking
12. Recurring invoices
13. Multi-currency support
14. Time entry editing by employees (with approval)
15. Client portal (clients can view their invoices)

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